

CHAPTER

2

THE SWEET SCIENCE

Insulin and Its Impact

So here's the good news: In this chapter, I'm going to go into some detail about how and why my program will help you. The bad news is that this chapter is more complicated than those that follow. It gets into the physiological purposes and actions of insulin.

While this chapter is crucial to understanding how my program works, it's not necessary that you grasp all of the biochemistry and physiology to follow my program. But I feel it's necessary to provide this information because it's based on solid science.

So feel free to skip this chapter and come back to it when you're more interested in WHY it will work rather than WHAT you should do. For the rest of you, put your science hat on and let's learn why my program works right now.

A MOMENT OF SCIENCE: HOW INSULIN WORKS

Insulin is necessary for living. This crucial hormone regulates blood sugar. Here's a step-by-step on how insulin works.

- Insulin is a hormone that is generated in the pancreas. It functions as a transport system.
- When you eat food, it is broken down through a series of steps, from chewing to digestive enzymes. By the time the food you've consumed enters your small intestine, the food is broken down into small enough molecular bits that they can be absorbed through your intestinal wall and into your blood stream.
- Once the nutrients have entered your blood stream, your pancreas delivers insulin to aid in delivery of these nutrients to where it perceives they are needed.

- Most people think insulin is ONLY triggered by consuming carbohydrates, but the truth is that insulin is released when we eat almost anything. Fiber and water are the only things we consume that do not trigger release of insulin. In fact consuming fiber mitigates the release of insulin because it slows entry of other foods into your system.
- Carbohydrates and sugary foods cause your blood-sugar levels to rise. When your blood-sugar levels are high, the pancreas kicks in, releasing insulin to bring your blood sugar down.
- Insulin moves the sugar in your blood to your liver and muscles to use blood glucose as a source of energy. But both your liver and muscles have a limited amount of storage, so any leftover blood sugar is then transported by insulin to be stored as bodyfat.
- Protein does not typically cause a spike in blood sugar, but it does (unexpectedly) cause a rise in insulin secretion. The insulin spike from eating protein lasts about the same amount of time as the spike from eating carbs. Even fats, like trans and saturated fats cause insulin release. So what does this mean?
- To boil it down, your body will not readily burn bodyfat until your body stops releasing insulin. You CANNOT burn stored fat for energy when your body is releasing insulin.

Every time we eat, our body goes through this process. Depending on the quantity and content of what you eat, insulin levels can remain elevated for as long as four hours after your meal. When insulin levels are elevated, your body burns glucose rather than stored fat, and your body can convert everything that enters your body to glucose, including protein and dietary fats.

INSULIN AND DIABETES: TYPE 1 AND TYPE 2

Insulin is necessary for living. This crucial hormone regulates blood sugar. Type 1 diabetics cannot produce insulin, and they must inject it frequently to survive, constantly monitoring their blood-sugar levels.

It's important that every single one of us understands the impact of insulin on our lives so that we make healthy nutrition choices. After all, we now have a type 2 diabetes epidemic in our country. The difference between type 1 diabetes and type 2 is this: Type 1 diabetics don't produce insulin, while type 2 diabetics don't

respond to insulin. Type 2 diabetes is a “lifestyle” disease, caused by poor nutrition habits and lack of exercise.

You may not believe this fact—it’s really shocking—but more than 100 million Americans suffer from type 2 diabetes or its precursors. That’s one out of three of everyone who lives in this country!

This epidemic is caused by what we choose to eat, when we consume it, and how much we take in. While type 1 diabetes must be controlled medically, type 2 diabetes is avoidable and/or reversible so long as you are not terribly ill from this lifestyle disease and you’re willing to follow a sound nutrition and workout regimen.

Following my guidelines will not only help you reduce bodyfat, but they will also help prevent this lifestyle disease.

INSULIN SENSITIVITY VS. RESISTANCE

Insulin sensitivity is the relationship between how much insulin is needed to lower blood sugars in your body with how much you need to drive healthy physiological digestive processes.

When you consume the proper foods, your body releases a modest amount of insulin. And when you take in these foods at the right times of day (with the proper amount of time between meals), insulin helps promote muscle building

when you train with weights without driving calories you’ve consumed to fat storage. If you’re following the rules I lay out in the next chapter, then your body will be insulin sensitive, helping you achieve your fat-loss and muscle-building goals.

SUPERHERO STORY: Carbs? Sure!

★ When the makeover people from BuzzFeed first sat down with me so I could explain my program to them they were a little surprised. They expected me to tell them horror stories of how I got Chris Evans in shape in 4 weeks after they saw the makeover piece I did with him on *EXTRA*. They expected they’d never be able to eat another carb and that they’d be performing five hours of cardio and training hard for another five hours every day.

After I went over everything with them they were completely relieved and excited at the same time. They believed they could achieve their goals and sustain them after we were done. To me, it’s all about being able to keep what you worked for and making your nutrition plan as easy as possible.

However, if you just consume whatever you want whenever you want, then you're setting your body up to become insulin resistant. This is the condition that creates type 2 diabetes and its precursors, which 100-plus million Americans already suffer from.

Insulin resistance is caused by consuming the wrong foods at the wrong time and in large quantities over a long period. Eventually, your cells stop responding to the normal flood of insulin you're unleashing by consuming unhealthy foods. Then, your pancreas responds by releasing even more insulin. This additional insulin doesn't drive calories to healthy processes, of course. Instead, it takes the food you're consuming to storage as bodyfat.

But, equally significantly, the large wash of insulin is training your muscle cells not to accept the benefits of insulin. This is particularly true if you're not stimulating these cells through exercise. This tandem of consuming the wrong foods at the wrong time in large quantities coupled with too little exercise is what causes people to develop type 2 diabetes.

Let me provide an example to better explain insulin resistance: Think of your body as a guy who never drinks, but then gets fairly drunk when he takes one stiff drink. And he likes it, so he starts drinking more and more. Eventually, he doesn't even feel the first drink, so he takes in more and more to get that buzz.

When you're taking in insulin-spiking foods every 2-3 hours, you're undermining the benefits of insulin while increasing the negative effects. You're shortening your "Burn Zone" (a term I'll explain in the next chapter), and you're simultaneously becoming more resistant to insulin's benefits.

WHAT'S THE PLAN? LEARN TO USE INSULIN

I've created my nutrition plan to help you take control of your insulin levels. Remember, insulin is a hormone that directly affects fat storage. If you learn how to monitor, regulate and control insulin, it can actually break down fat and build muscle for you.

It's not just about carbs. We have spent years believing carbs are the primary problem, but they are a catalyst, rather than the only culprit. My plan shows you how to mitigate hunger, break cravings, and build muscle all while burning stored body fat.

The key is to know how foods affect insulin and learn to reduce insulin

spikes. For instance, **if you eat three moderate meals of lean protein, unprocessed carbs, and fats, you will reduce the duration of your insulin spikes.**

Remember that.

FIND YOUR SET POINT

Everybody has a “set point,” a place their body naturally slides to in terms of the amount of calories they burn before additional intake goes to fat storage. It’s a challenge to move that set point and even tougher to keep it elevated.

Maybe you worked hard to lose five pounds, but then it seems like one night out is all it takes to pack back on those pounds. My program helps you change your set point, allowing you to maintain this new level where you burn fat more readily while holding and building muscle.

The first step is being absolutely vigilant about what you put in your body and when. Remember not all calories are created equal. If you eat 1,500 calories of ice cream a day, your body will look much different than if you eat 1,500 calories of eggs, fish, chicken, vegetables, and greens. Certain calories cause our bodies to store more bodyfat, while other calories cause certain hormones to secrete at higher levels than others.

I will prepare your body to get the most out of your workouts by placing you in the Burn Zone.

I’ll explain how you can utilize your bodyfat to power your day rather than relying on glucose, which not only keeps your fat in storage, but packs more in, regardless of how much or little you’re consuming.

When you’re taking in insulin-spiking foods every 2-3 hours, you’re undermining the benefits of insulin while increasing the negative effects. You’re shortening your Burn Zone, and you’re simultaneously becoming more resistant to insulin’s beneficial effects.

SUGAR IS YOUR ENEMY

Consuming even small amounts of processed sugar make it difficult to lose bodyfat. Red meat and sugar together can elevate your insulin levels for up to four hours after consumption. While red meat provides health advantages, you should avoid all foods with processed sugar. Here’s why sugar is the enemy.

Processed sugar, generally sucrose (table sugar) contains a large amount of glucose; it's half glucose and half fructose (fruit sugar). Neither of these types of sugar is healthy. Glucose causes blood hyperglycemia (high blood sugar in our body). Fructose can cause fatty liver and lead to insulin resistance, which is why high-fructose corn syrup is such a terrible additive to foods.

When you have large amounts of glucose in your blood, your pancreas sounds the alarm and releases a large amount of insulin to combat it. Insulin moves glucose into fat and muscle tissue. But insulin often overreacts to an influx of sugar, which causes your blood sugar to drop below normal levels, creating the condition called hypoglycemia.

This low level of blood sugar then makes you hungry, causing you to reach for unhealthy snacks and, even, healthy foods that you shouldn't consume at this time. But hypoglycemia particularly drives your craving for sugar or other fast-digesting carbs.

You feel good when you're eating sugar or other fast-digesting carbs, but soon afterwards you feel tired or hungry, and so you eat them again to feel more energized. And then this cycle repeats, over and over. That's what drives your body to pack on pounds of fat and feel more lethargic, never mind the long-term consequences such as diabetes, heart disease and cancer, all of which excess consumption of sugar is directly linked to. If you want to lose bodyfat and get healthier, then you need to cut out processed sugar, especially high-fructose corn syrup. End of story.

AVOID CRASH-AND-BURN DIETS

Here's another problem: Crash diets cause you to lose muscle mass, which depletes your ability to lose bodyfat. That's because you're reducing the effectiveness of muscle tissue to burn calories. Muscle cells are essentially tiny engines that burn calories, and so you want to increase them, not reduce them. That's true regardless of how much weight you're losing or how quickly. When you have less muscle, your body learns to conserve energy by slowing down your metabolism.

My mantra is the more lean muscle you have the higher your metabolism becomes because your body burns calories to maintain that muscle. Lose the muscle and your metabolism drops, making it more likely your body will hold onto (or add) bodyfat.

Maintaining or adding muscle is how you lose bodyfat. Encouraging insulin release at the proper times is crucial to this. After all, insulin is also an anabolic (muscle-building) hormone, and it stimulates protein synthesis.

I know I started this chapter by saying that elevated insulin levels are detrimental to burning fat. But it's equally important that you understand that insulin, when properly managed, is better than steroids at building muscle. In fact, many bodybuilders inject insulin at specific times of day to over-emphasize this effect. By the way, that's a TERRIBLE, potentially deadly, idea. Never do that.

But here is how you can get the most from beneficial insulin release. Think of this hormone as a transport system that drops off its passengers at three different locations, based on signaling. Your goal is to prioritize these through what and when you eat.

- 1) **Insulin drives nutrients to muscle tissue.** That's your goal when you're working out. Insulin supports recovery and muscle growth.
- 2) **Insulin drives nutrients to your liver.** This can be beneficial because it helps convert what you've consumed into what your body needs most at this time.
- 3) **Insulin drives nutrients to fat storage.** This is, exactly, what you want to avoid. The key is to avoid consuming foods that encourage insulin release when they don't support building muscle and fat loss.

INSULIN: USE IT, DON'T ABUSE IT

Since our bodies have this built in transport system, I want you to learn to use it to your advantage. You'll see this process at work in my BuzzFeed videos, such as "We Trained Like Superheroes For 30 Days," or "We lost 22% Body Fat In 6 Weeks," plus many others.

I supplement my program with branched-chain amino acids (BCAAs) from Hollywood Supps. BCAAs help promote muscle gain and fat loss. The reason I choose Hollywood Supps BCAAs is they are pure without other ingredients that will spike your insulin excessively.

Let me explain how this works. I have my clients take the recommended amount of BCAAs right before and after their workouts (I recommend leucine 1,600 mg, isoleucine 800 mg, valine 800 mg, all free form).

Timing is essential, as with everything else in my program. These amino

acids act as carriers that assist your muscles in synthesizing other amino acids needed for building muscle. BCAAs cause a deliberate slight spike in the release of insulin, which then pushes the amino acids straight into your muscles.

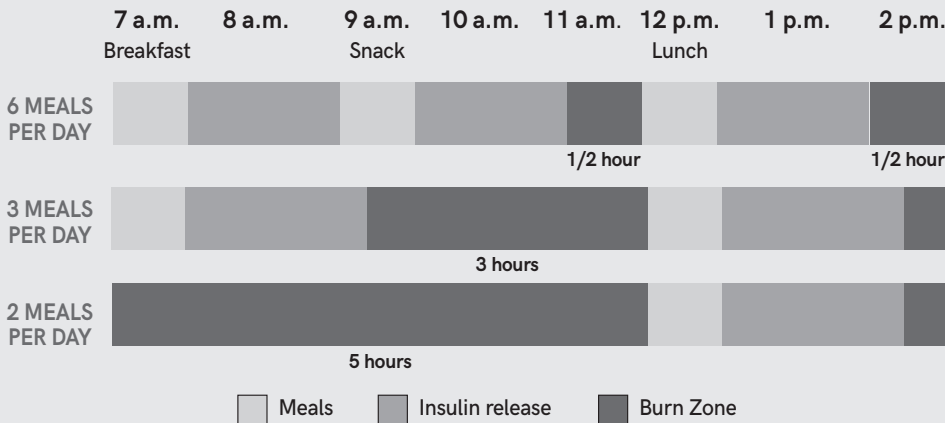
BCAAs are both anabolic and anticatabolic (they promote muscle gain and prevent protein breakdown/muscle loss). BCAAs cause a significant increase in protein synthesis and help in the release of growth hormone and insulin. They also help maintain good testosterone-to-cortisol ratios.

In a recent study published in the *Nutritional Journal of Medicine*, researchers concluded that catabolism of muscle tissue can be reduced by supplementing with BCAAs. The researchers concluded that if the anabolic phase is greater than the catabolic phase, then improvement in muscle strength and size will take place.

Other studies have shown that BCAAs cause exactly that and point directly to the reason why they are so effective in helping our bodies get lean. Muscle burns approximately 3 to 4 times the amount of calories that fat does just to

THE DAILY BURN ZONE:

Here's how you can understand the Burn Zone, and visualize when you're in it... and when you're not. Keep in mind that insulin typically spikes for about two hours after you eat a meal, so the Burn Zone doesn't begin until two hours after you complete your meal. You can eat as quickly or slowly as you like, so long as you understand this.



sustain itself. So the more lean muscle mass you have the more calories your body will burn just to sustain that muscle tissue.

This becomes a self-fulfilling prophecy: You tear down muscle tissue when you work out with weights. You do not build muscle when you are lifting; you are actually microscopically tearing muscle. Then, the moment you stop working out your body starts to repair itself and build the muscle. So the nutrition you consume after workouts provides the building blocks you need to add new muscle tissue.

WHAT YOU SHOULD DO

What if I told you that I've discovered a new method of burning fat while eating fewer meals a day? A way to build muscle more readily than ever before while also reducing bodyfat? In addition, you won't have to spend hours on cardio machines. When you follow my Superhero Nutrition program everything falls into place.

All you have to do is follow the 10 rules I explain in the next chapter.

Remember to add in 2 Tbsp. of raw unfiltered apple cider vinegar before each meal to shorten your insulin release window by about 30%. That will significantly increase your Burn Zone, particularly when you're consuming two or three meals per day. When you're consuming 6 meals a day, shortening insulin release is less significant because you're often already eating another meal before you reach the Burn Zone.

